(a) Cheap: measured on the doser itself
measured (13 powders)
proposed
infer the properties of a new powder
known properties speed up calibration
Feed factor
mg per auger revolution at three tilts; before and after humid air
bulk
tapped
tap
closed
“Ghost dose”
transparent doser, outlet closed, tapped: bulk and tapped density
θ
Angle of repose
a camera on the pile the doser pours
(b) Physics-based ML
surrogate models and DEM simulations
Relational matrix
Hall
PSD
Shear
Tap
Moisture
Feed factor
Ghost dose
Angle of repose
expected
to be tested
same quantity
Powder state and history
composition, and how it was handled
made
dried
reused
sieved
stored
(c) Standard tests: a separate instrument for each, some only in outside labs
Hall/Carney flow
50 g through a funnel ASTM B213, B964
in-house
Particle size (PSD)
SEM (BSE) images or laser diffraction
in-house or outside lab
Shear cell, ff_c
flowability under load ASTM D6773, D7891
outside lab
Tapped density
graduated cylinder ASTM B527
in-house
Moisture
loss on drying: weigh, dry, reweigh
in-house